

Structural Preservation & Graceful Degradation

Why the `<picture>` element requires NO JavaScript polyfills

Legacy Parsers (e.g., IE 11)

✖ Unknown Elements Ignored

Legacy parsers do not recognize `<picture>` or `<source>`. Per HTML5 spec, they treat unknown tags as generic inline elements and safely **parse their children**. No JS errors are thrown; the structural hierarchy is maintained.

✔ Standard Baseline Reached

The parser continues down and natively hits the `<img>` tag. Recognizing this standard element, it fires a request for `fallback.jpg` and renders it perfectly.

Modern Parsers (Chrome, Safari)

⚙ Context Wrapper Initialized

Modern parsers recognize `<picture>` not as a visual element, but as a **wrapper**. It evaluates the `<source>` tags for media queries and format support, picking the best match.

🔗 Property Mutation (`currentSrc`)

The `<picture>` element **renders nothing**. Instead, it passes the URL of the winning `<source>` directly to the `<img>` tag, mutating its `currentSrc` property.

```
index.html
1 <picture>
2   <source type="image/avif" srcset="hero.avif">
3   <source media="(min-width: 800px)" srcset="large.jpg">
4   
5 </picture>
```

📌 The `<img>` Tag is the Absolute Structural Anchor

Without the `<img>` tag, a `<picture>` element is entirely invisible in **both** modern and legacy browsers. The `<picture>` tag provides the cascading logic, but the `<img>` tag paints the actual pixels to the DOM.